

COMPSYS 306 Project 2 – Week 9 Data Collection Summary

Group: Group 20

1. Number of Classes and Samples

The collected dataset contains 6 classes with a total of 1966 images.

Class	Number of Images
Stop	214
Speed	225
Blue light	235
Yellow light	224
Sheep	284
Road	784
Total	1966

The Road class represents normal track images where no target sign, traffic light, or sheep is visible.

2. Diversity of Images

Images were collected under varied viewing conditions to improve the diversity of the dataset. Each target object (Stop, Speed, Blue light, Yellow light and Sheep) was captured with variation in the following:

- Robot-to-sign distance during capture
- Horizontal position of the object within the camera view
- Orientation of the object relative to the robot

The Road class contains normal driving scenes including straight sections, left and right curves, and varied positions of the black line in the camera view.

3. Image Resolution

The Stop, Speed, Blue light, Yellow light and Sheep images were stored at 64×48 pixels. The Road images collected during motion capture were stored at 320×240 pixels.

The Road images will be resized to 64×48 pixels during the preprocessing stage so that all images have a consistent input size for model training and to reduce computational resource usage.